

Prof. Samir Kenouche

Quantum Chemistry and Mathematical Chemistry
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ORCID: orcid.org
Place of Birth: Béjaïa | 3 children (2 daughters, 1 son)

My research focuses on developing non-local theoretical descriptors within Conceptual Density Functional Theory (c-DFT). Recent contributions include:

- **S. Kenouche**. The Dual Delocalization Descriptor: First- and Second-Order Responses of Bond-Resolved Electron Sharing. *Physical Chemistry Chemical Physics - RSC*, 28(31), 19133–19145. https://doi.org/10.1039/D6CP01777K
- **S. Kenouche**. Spin-Resolved Nonlocal Second-Order Fukui Index for Open-Shell Systems from Delocalization Index Response. *The Journal of Physical Chemistry A - ACS*, 130(27), 5190–5204. https://pubs.acs.org/doi/10.1021/acs.jpca.6c01997
- **S. Kenouche** and J. Ignacio Martínez-Araya. Electron-Number Response of QTAIM Delocalization Indices: A Differential Decomposition into Population and Orbital-Relaxation Contributions. *The Journal of Physical Chemistry A - ACS*. https://doi.org/10.1021/acs.jpca.6c05379
- **S. Kenouche**. Analytical derivation of the non-local Second-Order Fukui Index via the first-order response of the delocalization index. *Computational and Theoretical Chemistry - Elsevier*, 1259(5), 115748. https://doi.org/10.1016/j.comptc.2026.115748
- **S. Kenouche**. Computational framework for non-local Second-Order Fukui Indices (SOFI) based on atomic overlap integrals: Interorbital coupling and self-delocalization effects. *Computational and Theoretical Chemistry - Elsevier*, 1264, 115949. 10.1016/j.comptc.2026.115949
- **S. Kenouche**. Orbital-resolved decomposition of the delocalization index in closed- and open-shell systems. *Theoretical Chemistry Accounts - Springer*, 145(75), 1-15. https://doi.org/10.1007/s00214-026-03328-3
- **S. Kenouche**. Spectral analysis of the QTAIM localization–delocalization matrix: Electronic localization and delocalization modes in aromatic systems. *Structural Chemistry - Springer*. https://doi.org/10.1007/s11224-026-02897-0
- **S. Kenouche** and J. Ignacio Martínez-Araya. An explicit density-matrix formulation for spin-resolved local reactivity in open-shell systems. *Computational and Theoretical Chemistry - Elsevier*, 1261, 115826. DOI:10.1016/j.comptc.2026.115826
- **S. Kenouche** and J. Ignacio Martínez-Araya. The linear response function $\chi(\mathbf{r}, \mathbf{r}')$: another perspective. *Journal of Mathematical Chemistry - Springer*, 62(11), 2880–2888. https://doi.org/10.1007/s10910-024-01578-9

Software Implementation & Research Impact — Multiwfn (v2026.10.1)

- **Official Software Integration:** My analytical theory published in *PCCP* (S. Kenouche. *Physical Chemistry Chemical Physics*, 28(31), 19133–19145, https://doi.org/10.1039/D6CP01777K) has been officially integrated into the Conceptual DFT module of **Multiwfn** (the world's most widely used quantum chemistry post-processing code) by Prof. Tian Lu.
- **Official Release Log Entry:**

  > *"[2026.9.23] The dual delocalization descriptor (DDD) proposed in Phys. Chem. Chem. Phys., 28, 19133 (2026) by Samir Kenouche has been implemented in conceptual DFT module. This theory analytically quantifies responses of the interatomic delocalization index (DI) to variations in the total electron number. See Section 3.25.6 of manual for introduction and Section 4.22.6 for example."*

- **Documentation & Examples:** Authored the theoretical foundations and worked benchmark examples incorporated into Sections 3.25.6 and 4.22.6 of the official Multiwfn manual.
- **Official Link:** Multiwfn Update History & Download Page

Current Professional Position

- Since February 2026: Promoted to Professor - University of Biskra.
- Since May 2021: Senior Lecturer "A" - University of Biskra.

Degrees

- May 2021: Habilitation à Diriger des Recherches (HDR). Topic: **Conceptual DFT and Quantitative Analysis of Weak Interactions** (Quantum Chemistry).
- 2009 – 2013: *PhD in Physics*, University of Montpellier. Title: **Experimental Studies and Modeling of the Distribution Dynamics of Contrast Agents in NMR Imaging**. PDF
- 2007 – 2009: Magister's Degree (Advanced Master's Degree, 6-year program), M. Mammeri University of Tizi-Ouzou. *Specialization in Solid-State Physics and Chemistry*.
- June 2006: *DES in Physical Chemistry*, A. Mira University of Béjaia.

Supervision Activities

Master's Level

- 2011-2012: University of Montpellier, supervised Two Master's Degrees (Condensed Matter Physics).
- 2014-2015: University of Béjaia, supervised Three Master's Degrees (Biophysics and Imaging).
- 2015-2026: University of Biskra, supervised about a dozen Master's Degrees (Quantum Chemistry).
- 2022-2023: Start-up Project, development of an interactive interface for Quantum Chemistry calculations.

PhD Supervision

- 2021-2024: PhD thesis of *Bachir Nassima*: **Searching for Neutralisers of Energetic Organic Compounds: A Theoretical Approach from Quantum Chemistry** (supervisor).
- 2019-2022: PhD thesis of *Belkadi Ahlem*: **Investigation of Cytotoxic Properties of Some Heterocyclic Derivatives by Molecular Modeling Approaches** (co-supervisor).
- 2018-2022: PhD thesis of *Djebaili Rachida*: **Structure-Activity Modeling Approaches for Prediction of Chemical Reactivity of Some Heterocyclic Compounds** (co-supervisor).

Teaching Activities

- 2014-2015: **University of Béjaia**, course on Radiation-Matter Interactions and Imaging.
- 2014-2015: Private School "Les Colombs", Physics-Chemistry (Final year of Science track, French Baccalaureate - Béjaia).
- 2015-2026: **University of Biskra**, courses: Numerical Methods and Programming (L2), Mathematical Methods and Algorithms (M1), Quantum Chemistry (M1), Atomic and Molecular Spectroscopy (M1), Lab: Artificial Intelligence and Machine Learning (M1).

International Publications

- 2026 : **S. Kenouche**. *Physical Chemistry Chemical Physics - RSC*, 28(31), 19133–19145. https://doi.org/10.1039/D6CP01777K
- 2026 : **S. Kenouche**. *The Journal of Physical Chemistry A - ACS*, 130(27), 5190–5204. https://pubs.acs.org/doi/10.1021/acs.jpca.6c01997
- 2026 : **S. Kenouche** and J. Ignacio Martínez-Araya. *The Journal of Physical Chemistry A - ACS*. https://doi.org/10.1021/acs.jpca.6c05379
- 2026 : **S. Kenouche**. *Computational and Theoretical Chemistry - Elsevier* 1259 (5), 115748. https://doi.org/10.1016/j.comptc.2026.115748
- 2026 : **S. Kenouche** and J. Ignacio Martínez-Araya. *Computational and Theoretical Chemistry - Elsevier*, 1261, 115826. DOI:10.1016/j.comptc.2026.115826
- 2026 : **S. Kenouche**. *Computational and Theoretical Chemistry - Elsevier*, 1264, 115949. 10.1016/j.comptc.2026.115949
- 2026 : **S. Kenouche**. *Theoretical Chemistry Accounts - Springer*, 145(75), 1-15. https://doi.org/10.1007/s00214-026-03328-3
- 2026 : **S. Kenouche**. *Structural Chemistry - Springer*. https://doi.org/10.1007/s11224-026-02897-0
- 2025 : J. Ignacio Martínez-Araya, **S. Kenouche**, F. Barrios, T. Lu, H. Kawabata. *Computational and Theoretical Chemistry - Elsevier* 1252 (10), 115328. https://doi.org/10.1016/j.comptc.2025.115328.
- 2025 : I. Bensahbane, N. Melkemi, I. Daoud, **S. Kenouche**. *ChemistrySelect - Wiley* 11 (5), e01897. https://doi.org/10.1002/slct.202501897.
- 2025 : S. Rahali, A. Zaiter, R. Ben Said, **S. Kenouche**, Y. Belhocine et al. *Journal of Cluster Science - Springer* 36 (153), 1-10. https://doi.org/10.1007/s10876-025-02875-2.
- 2024 : **S. Kenouche**, J. Ignacio Martínez-Araya. *Journal of Mathematical Chemistry - Springer* 62 (11), 2880-2888. https://doi.org/10.1007/s10910-024-01578-9.

- 2024 : **S. Kenouche**, N. Bachir, W. Bouchel, J. Ignacio Martínez-Araya. *Journal of Molecular Graphics and Modelling - Elsevier* 129 (2), 108728. https://doi.org/10.1016/j.jmgm.2024.108728.
- 2024 : N. Bachir, **S. Kenouche**, J. Ignacio Martínez-Araya. *Journal of Molecular Graphics and Modelling - Elsevier* 126, 108645-108655. https://doi.org/10.1016/j.jmgm.2023.108645.
- 2023 : **S. Kenouche**, N. Bachir, J. Ignacio Martínez-Araya. *ChemPhysChem - Wiley* 24 (2), e202200488. https://doi.org/10.1002/cphc.202200488.
- 2023 : N. Bachir, **S. Kenouche**, J. Ignacio Martínez-Araya. *Journal of Molecular Graphics and Modelling - Elsevier* 118 (1), 108341-108351. https://doi.org/10.1016/j.jmgm.2022.108341.
- 2023 : A. Zekri, D. Harkati, **S. Kenouche**, B. Saleh, R. Alnajjar. *Journal of Biomolecular Structure and Dynamics - Taylor* 41 (20), 11078-11100. https://doi.org/10.1080/07391102.2022.2159877.
- 2023 : R. Kherachi, I. Daoud, N. Melkemi, **S. Kenouche** et al. *Biologia - Springer*. https://doi.org/10.1007/s11756-023-01528-x.
- 2023 : M. Mettai, I. Daoud, F. Mesli, **S. Kenouche**, N. Melkemi, A. Belkadi. *In Silico Pharmacology - Springer* 11 (3), 1-22. https://doi.org/10.1007/s40203-023-00139-3.
- 2022 : **S. Kenouche**, J. Ignacio Martínez-Araya. *Journal of Molecular Graphics and Modelling - Elsevier* 116 (51), 108273-108284. https://doi.org/10.1016/j.jmgm.2022.108273.
- 2022 : **S. Kenouche**, C. Sandoval, J. Ignacio Martínez-Araya. *Chemical Physics Letters - Elsevier* 801 (6), 139708-139716. https://doi.org/10.1016/j.cplett.2022.139708.
- 2022 : A. Belkadi, **S. Kenouche**, N. Melkemi, I. Daoud, R. Djebaili. *Struct. Chem. - Springer* 33 (3), 833–858.
- 2022 : R. Djebaili, **S. Kenouche**, I. Daoud, N. Melkemi et al. *Struct. Chem. - Springer* 11 (9), 1-33. https://doi.org/10.1007/s11224-022-02029-4.
- 2021 : **S. Kenouche**, A. Belkadi, R. Djebaili, N. Melkemi. *Journal of Molecular Graphics and Modelling - Elsevier* 104 (5), 107828-107840. https://doi.org/10.1016/j.jmgm.2020.107828.
- 2021 : A. Belkadi, **S. Kenouche**, N. Melkemi, I. Daoud, R. Djebaili. *Struct. Chem. - Springer* 32 (6), 2235–2249. https://doi.org/10.1007/s11224-021-01796-w.
- 2020 : A. Zekri, D. Harkati, **S. Kenouche** and B. A/Saleh. *Journal of Molecular Structure - Elsevier* 1217 (6), 128442-128454. https://doi.org/10.1016/j.molstruc.2020.128442.
- 2018 : **S. Kenouche**, D. Harkati, M. Ghamri, A/R. Chikhaoui and N. Melkemi. *Journal of Computational Chemistry & Molecular Modelling* 2 (3), 1-10.
- 2016 : E.M. Halidi, E. Nativel, M. Akel, **S. Kenouche**, C. Coillot, E. Alibert, R. Schimpf, M. Zanca, P. C. Stein and C. Goze-Bac. *PLoS ONE: Electromagnetism* 11 (1), 1-8. https://doi.org/10.1371/journal.pone.0144483.
- 2015 : M. Perrier, A. Gallud, A. Ayadi, **S. Kenouche**, C. Porredon, M. Gary-Bobo, J. Larionova, C. Goze-Bac, M. Zanca, M. Garcia, I. Basile, J. Long, M. Borras and Y. Guari. *Nanoscale - ACS* 7 (28), 11899-11903. https://doi.org/10.1039/C5NR01557J.
- 2014 : **S. Kenouche**, J. Larionova, N. Bezzi, Y. Guari, M. Cieslak, M. Zanca, L. Lartigue, N. Bertin, C. Godin and C. Goze-Bac. *Powder Technology - Elsevier* 255 (10), 60-65. https://doi.org/10.1016/j.powtec.2013.07.038.

- 2014 : **S. Kenouche**, M. Perrier, N. Bertin, J. Larionova, A. Ayadi, M. Zanca, J. Long, N. Bezzi, P. Stein, Y. Guari, M. Cieslak, C. Godin and C. Goze-Bac. *Magnetic Resonance Imaging - Elsevier* 32 (10), 1418-1427. https://doi.org/10.1016/j.mri.2014.08.005.
- 2013 : M. Perrier, **S. Kenouche**, J. Long, T. Kalaivani, J. Larionova, C. Goze-Bac, A. Lascialfari, N. Baril, C. Guérin, B. Donnadieu, A. Trifonov and Y. Guari. *Inorg. Chem. ACS* 52 (23), 13402-13414. https://doi.org/10.1021/ic401710j.
- 2012 : M. Cieslak, F. Boudon, **S. Kenouche**, M. Zanca, C. Goze-Bac, M. Génard, C. Godin and N. Bertin. *Acta Horticulturae* 957 (27), 239-245. DOI:10.176580/ActaHortic.2015.1068.6.
- 2012 : M. Cieslak, M. Génard, **S. Kenouche**, C. Goze-Bac, C. Godin and N. Bertin. *Acta Horticulturae* 1068 (6), 59-66. DOI:10.17660/ActaHortic.2012.957.27.

National and International Presentations

- From 2011 to 2024: I have delivered around twenty national and international presentations in the field of Computational and Quantum Chemistry.

Books

- December 2022: **S. Kenouche**, *Numerical Methods and Programming with Matlab*, OPU, ISBN 978.9961.0.2411.9
- June 2023: **S. Kenouche**, *Mathematical Methods for Physics*, OPU, ISBN 978.9961.0.2443.0

Research Projects

- 2022-2026: PRFU Project. **Search for Neutralizers of Energetic Organic Compounds Using Quantum Calculatons** (Code B00L01UN070120220004).
- 2018-2022: PRFU Project. **Predictive Models and Structur-Activity Relationships for Anti-cancer Activity** (Code B00L01UN070120180002).
- 2013-2015: CNEPRU Project. **Semiconductor Surfaces for Photocatalytic Degradation** (Code NE00620130006).

PhD Entrance Exams (LMD System)

- 2022-2023: Design of exam topics, Group Theory module
- 2019-2020: Design and grading, Group Theory module
- 2018-2019: Design and grading, Quantum Chemistry module

Computer Skills

Programming Languages:	Matlab, R, Maple, Python	**Scientific Tools:**	LaTeX, Beamer
Operating Systems:	Unix/Linux, Windows	**Software:**	MatNMR, ImageJ
Office Tools:	OpenOffice, Microsoft Office	**Others:**	Qtiplot, Sigmaplot
Web:	HTML, CSS, PHP	**Graphics:**	SVG, Inkscape

Languages

- Kabyle: Native
- French: Fluent
- Arabic: Fluent
- English: Scientific proficiency

Scientific Events

- 2020: Scientific Committee Member. *13th International Days of Theoretical and Computational Chemistry*.
- 2020: Co-Chair of the Organizing Committee. *13th International Days of Theoretical and Computational Chemistry*.

Interdisciplinary Training

- April 2013: Image Processing (18h), Campus Saint Priest Montpellier. *Trainer: William Puech*.
- June 2012: Communicating Scientific Topics to a Non-Expert Audience (20h). *Trainer: Agnès Seye*.
- May 2012: Introduction to Pedagogical Tools for Higher Education (21h). *Trainer: Sébastien Balme*.
- April 2011: Statistics for Experimenters (20h). *Trainer: Gilles Ducharme*.
- January 2011: Qualitative Physics (16h). *Trainer: Michel Dyakonov*.

DONE IN BISKRA, OCTOBER 20, 2026
SAMIR KENOUCHE

